



**Rajshahi University of Engineering & Technology
(RUET)**

Department of Electrical and Computer Engineering (ECE)

**Course Name: Electrical Machine-I Sessional
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Lab Report - 1

**Experiment Name: Anatomy of Three-Phase AC
Induction Motor**

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1 Objectives

The primary objective of this experiment is to dismantle, examine, and understand the internal and external physical construction (anatomy) of a three-phase AC induction motor. By the end of the experiment, we aim to identify each main component and understand its specific function within the machine.

2 Apparatus Required

- A three-phase AC induction motor (completely assembled or prepared for disassembly)
- Mechanical tool kit (wrenches, screwdrivers, pliers, mallet, and bearing puller)
- Multimeter (for basic continuity checks)
- Safety equipment (gloves, safety goggles)
- Cleaning cloths and lubricant

3 Theory of Three-Phase Induction Motor

The three-phase AC induction motor is the most commonly used industrial electric motor due to its robust construction and simple design. It operates on the principle of electromagnetic induction (Faraday's Law).

When a three-phase AC supply is applied to the stator winding, it creates a Rotating Magnetic Field (RMF). This RMF sweeps across the stationary rotor conductors, inducing an Electromotive Force (EMF) and consequently a current within them. According to Lorentz force principles, the interaction between the induced rotor current and the stator's magnetic field generates torque, causing the rotor to spin in the direction of the magnetic field.

The synchronous speed (N_s) of the rotating magnetic field is determined by the frequency (f) of the supply and the number of poles (P), given by the formula:

$$N_s = \frac{120f}{P}$$

4 Construction (Anatomy) of the Induction Motor

4.1 Stator

The stator is the stationary part of the motor. It is built from high-grade alloy steel laminations to reduce eddy current losses. It provides mechanical support and a magnetic path for the machine.

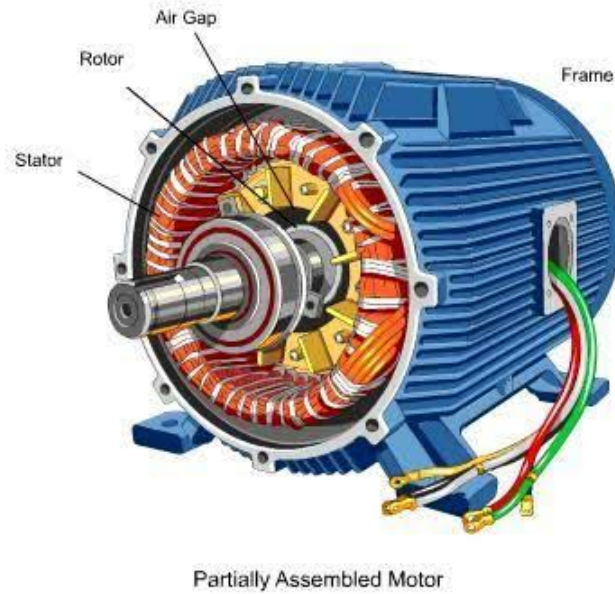


Figure 1: Assembled three-phase induction motor used in the experiment.



Figure 2: Stator core and its slots.

4.2 Rotor

The rotor is the rotating core of the motor. For this experiment, we observed a squirrel-cage rotor, which consists of a laminated cylindrical core with parallel slots holding heavy copper or aluminum bars, short-circuited by end rings.



Figure 3: Squirrel-cage rotor.

4.3 Shaft

The shaft is a solid steel rod running through the center of the rotor. It transmits the mechanical power generated by the motor to the external load.



Figure 4: Rotor shaft.

4.4 Stator Windings

Copper coils are placed inside the stator slots. These windings are connected in either a star or delta configuration and are responsible for producing the rotating magnetic field when energized.

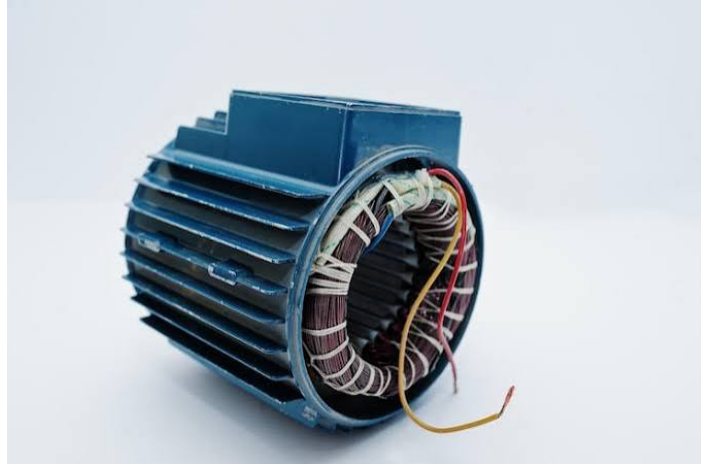


Figure 5: Three-phase stator windings in the slots.

4.5 Bearings

Bearings are fitted onto both ends of the shaft. They minimize friction and support the rotor, keeping it perfectly centered within the stator.



Figure 6: Ball bearing supporting the shaft.

4.6 Air Gap

The air gap is the tiny, uniform physical distance between the stator inner radius and the rotor outer radius. A smaller air gap improves the magnetic flux transfer and motor efficiency.

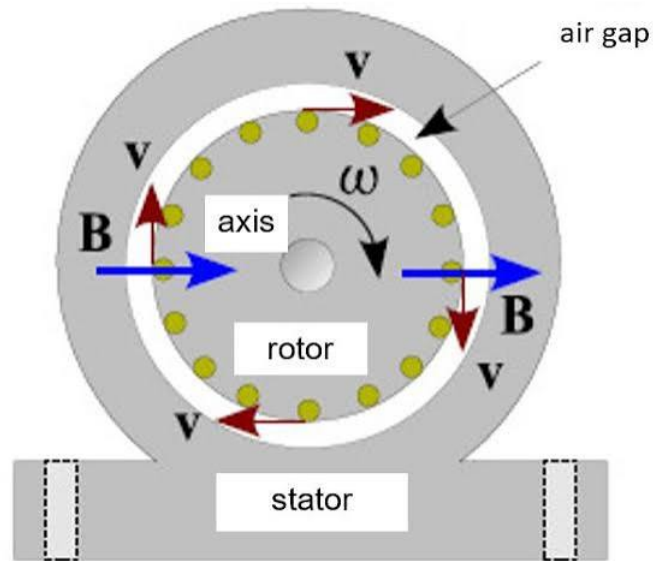


Figure 7: Air gap between the stator and rotor.

4.7 Cooling Fan

Mounted on the non-drive end of the shaft, the cooling fan spins with the rotor, blowing ambient air over the motor's outer frame to dissipate heat.



Figure 8: Shaft-mounted cooling fan.

4.8 Terminal Box

Located on the outside of the motor housing, the terminal box houses the connection points where the external three-phase power supply meets the internal stator winding leads.

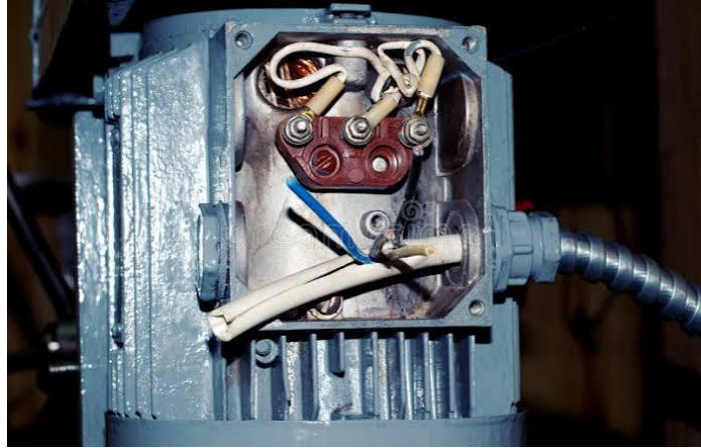


Figure 9: Terminal box with the winding leads.

4.9 Frame (Housing)

The outer frame, often cast iron or aluminum, encloses the entire motor. It features external cooling fins that increase the surface area to aid in thermal dissipation.



Figure 10: Outer frame (housing) with cooling fins.

4.10 End Shields (End Covers)

These cover the ends of the motor frame. They house the bearing assemblies and help keep the rotor precisely aligned with the stator.



Figure 11: End shield (end cover) housing a bearing.

5 Procedure

1. **Safety First:** Verified that the motor was completely disconnected from any electrical power source.
2. **External Inspection:** Photographed and noted the external components (Frame, Terminal Box, Cooling Fan cover).
3. **Dismantling:** Used appropriate tools to carefully remove the fan cover, followed by the cooling fan itself.
4. **Removing End Shields:** Unscrewed the bolts holding the end shields and gently tapped them off using a soft mallet.
5. **Extracting the Rotor:** Carefully pulled the rotor assembly (along with the shaft and bearings) out of the stator core without scraping the inner stator windings.
6. **Observation:** Examined the stator slots, windings, and squirrel cage rotor bars.
7. **Reassembly:** Reassembled all parts in the reverse order of disassembly, ensuring the bearings were properly seated and the rotor spun freely.

6 Observation

Upon opening the motor, we observed that the stator comprises thin steel laminations packed tightly together with copper windings secured in the slots. The rotor was of the squirrel-cage type, evident from the solid end rings with no slip rings or external

connections. The air gap was notably narrow and uniform. The bearings were well-greased, allowing smooth rotation of the shaft.

7 Result

The three-phase AC induction motor was successfully disassembled and reassembled. All major structural components—including the stator, rotor, shaft, windings, bearings, and housing—were accurately identified, and their physical functions were studied in line with theoretical concepts.

8 Discussion

The physical inspection revealed why squirrel-cage induction motors are so widely used. The absence of brushes, commutators, or slip rings on the rotor makes it nearly maintenance-free and highly rugged. We also noted the importance of the cooling fins and fan; because the motor generates significant heat during induction, the large surface area of the fins is vital to prevent insulation failure in the stator windings.

9 Precautions

- Ensured all power sources were strictly isolated before touching the motor.
- Handled the heavy rotor carefully to prevent dropping it or damaging the delicate stator windings during extraction.
- Kept all nuts, bolts, and small components organized in a tray to avoid misplacement during reassembly.
- Used a soft mallet rather than a metal hammer to avoid cracking the cast iron end shields.

10 Conclusion

This hands-on experiment provided a clear, practical understanding of a three-phase AC induction motor's anatomy. By bridging theoretical principles with physical component observation, we verified how the stator's stationary structure and the rotor's robust design work together to efficiently convert electrical energy into mechanical energy.

References

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